THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1%	Loss % = (p²/100) = 10²/100
2	**(a)** 60500	50000×(1+10/100)²
3	**(c)** 40500	50000×(1−10/100)²
4	**(a)** −4%	(1+20/100)(1−20/100) − 1
5	**(b)** 2500	A = P(1+20/100)², CI = A − P
6	**(a)** 26%	(1+40/100)(1−10/100) − 1
7	**(b)** 7500	Difference = P(R/100)²
8	**(d)** 50%	SI = 2P so 2P = P×R×T/100 and R = 200/T
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{25}{100} \times 360 = 90$; R is the rule used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$800 - 80 = 720$, so A is true. R is also true but concerns profit and loss, not the discount, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Final = (1+25/100)(1−10/100) (1) = 112.5% of original, so more by 12.5% (1).	1+1
12	Marked price = ₹ 2100 (1); SP = ₹ 1890 (1).	1+1
13	A = 2500(120/100)² = ₹ 3600 (1); A/P = 1.44 (1).	1+1
14	Final = (1+40/100)(1−30/100) (1) = 98% of original, so less by 2% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	CP1 = 1980×100/110 = 1800 (1); CP2 = 1980×100/90 = 2200 (1); total CP − total SP = ₹ 40, a loss (1).	1+1+1
16	V = P(1−r/100)³ (1) = 500000×90/100 three times (1) = ₹ 364500 (1).	1+1+1
17	MP = ₹ 1200 (1); SP = ₹ 960 (1); gain/loss % = 20% (1).	1+1+1
	OR: CP1 = 2640×100/120 = 2200 (1); CP2 = 2640×100/80 = 3300 (1); total CP − total SP = ₹ 220, a loss (1).	1+1+1
18	V = P(1−r/100)³ (1) = 400000×80/100 three times (1) = ₹ 204800 (1).	1+1+1
	OR: MP = ₹ 1560 (1); SP = ₹ 1248 (1); gain/loss % = 4% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 20% of $2500 = ₹500$ (1). (ii) $2500 - 500 = ₹2{,}000$ (1). (iii) 12% of $2000 = ₹240$ (1). (iv) Final amount $= 2000 + 240 = ₹2{,}240$ (1); it is $₹2500 - 2240 = ₹260$ less than the marked price (1).	1+1+1+2
20	Year 1: A1 = ₹ 55000, interest ₹ 5000 (1); year 2: A2 = ₹ 60500, interest ₹ 5500 (2); year 3: A3 = ₹ 66550, interest ₹ 6050 (1); total ₹ 16550 = A3 − P (1).	1+2+1+1
	OR: Let CP = x. SP = x(1+60/100)(1−25/100) = 1.2x (2); profit = 0.2x = 100 (2); x = ₹ 500 (1).	2+2+1

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 51200	80000×(1−20/100)²
2	**(c)** −17.5%	(1+10/100)(1−25/100) − 1
3	**(a)** 4000	A = P(1+10/100)², CI = A − P
4	**(a)** 20%	(1+50/100)(1−20/100) − 1
5	**(d)** 15000	Difference = P(R/100)²
6	**(b)** 25%	SI = 2P so 2P = P×R×T/100 and R = 200/T
7	**(d)** 2.25%	Loss % = (p²/100) = 15²/100
8	**(d)** 90000	62500×(1+20/100)²
9	**(c)** A is true but R is false.	Amount $= 1000 \times 1.1^2 = 1210$, so CI $= 210$ and A is true. SI would be ₹200, so R is false.
10	**(d)** A is false but R is true.	SP $= 500 \times 0.8 = 400$, so A is false; R is a true statement.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Marked price = ₹ 3000 (1); SP = ₹ 2400 (1).	1+1
12	A = 4000(110/100)² = ₹ 4840 (1); A/P = 1.21 (1).	1+1
13	Final = (1+15/100)(1−10/100) (1) = 103.5% of original, so more by 3.5% (1).	1+1
14	Marked price = ₹ 5400 (1); SP = ₹ 3780 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	CP1 = 1950×100/125 = 1560 (1); CP2 = 1950×100/75 = 2600 (1); total CP − total SP = ₹ 260, a loss (1).	1+1+1
16	V = P(1−r/100)³ (1) = 250000×80/100 three times (1) = ₹ 128000 (1).	1+1+1
17	MP = ₹ 3200 (1); SP = ₹ 2240 (1); gain/loss % = 12% (1).	1+1+1
	OR: CP1 = 3360×100/140 = 2400 (1); CP2 = 3360×100/60 = 5600 (1); total CP − total SP = ₹ 1280, a loss (1).	1+1+1
18	V = P(1−r/100)³ (1) = 800000×90/100 three times (1) = ₹ 583200 (1).	1+1+1
	OR: MP = ₹ 650 (1); SP = ₹ 585 (1); gain/loss % = 17% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $40 \times 25 =$ ₹1,000 (1). (ii) $30 \times 35 =$ ₹1,050 (1). (iii) $10 \times 20 =$ ₹200 (1). (iv) Total SP = ₹1,250; profit $= 1250 - 1000 =$ ₹250 (1); profit $\% = \frac{250}{1000} \times 100 = 25\%$ (1).	1+1+1+2
20	Year 1: A1 = ₹ 44000, interest ₹ 4000 (1); year 2: A2 = ₹ 48400, interest ₹ 4400 (2); year 3: A3 = ₹ 53240, interest ₹ 4840 (1); total ₹ 13240 = A3 − P (1).	1+2+1+1
	OR: Let CP = x. SP = x(1+80/100)(1−30/100) = 1.26x (2); profit = 0.26x = 260 (2); x = ₹ 1000 (1).	2+2+1

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
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- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6400	A = P(1+25/100)², CI = A − P
2	**(c)** 20%	(1+60/100)(1−25/100) − 1
3	**(a)** 4000	Difference = P(R/100)²
4	**(a)** 20%	SI = 2P so 2P = P×R×T/100 and R = 200/T
5	**(a)** 9%	Loss % = (p²/100) = 30²/100
6	**(a)** 125000	80000×(1+25/100)²
7	**(d)** 36100	40000×(1−5/100)²
8	**(d)** −10%	(1+50/100)(1−40/100) − 1
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{2000 \times 5 \times 3}{100} = 300$; R is the formula used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Amount $= 5000 \times 1.21 = 6050$, CI $= 1050$, so A is true. R is true but does not give the value of the two-year interest.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A = 1600(125/100)² = ₹ 2500 (1); A/P = 1.56 (1).	1+1
12	Final = (1+60/100)(1−50/100) (1) = 80% of original, so less by 20% (1).	1+1
13	Marked price = ₹ 3200 (1); SP = ₹ 2400 (1).	1+1
14	A = 1000(130/100)² = ₹ 1690 (1); A/P = 1.69 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	CP1 = 990×100/110 = 900 (1); CP2 = 990×100/90 = 1100 (1); total CP − total SP = ₹ 20, a loss (1).	1+1+1
16	V = P(1−r/100)³ (1) = 160000×75/100 three times (1) = ₹ 67500 (1).	1+1+1
17	MP = ₹ 1800 (1); SP = ₹ 1350 (1); gain/loss % = -10% (1).	1+1+1
	OR: CP1 = 1200×100/120 = 1000 (1); CP2 = 1200×100/80 = 1500 (1); total CP − total SP = ₹ 100, a loss (1).	1+1+1
18	V = P(1−r/100)³ (1) = 625000×80/100 three times (1) = ₹ 320000 (1).	1+1+1
	OR: MP = ₹ 1620 (1); SP = ₹ 972 (1); gain/loss % = 8% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{8000 \times 10 \times 1}{100}$ = ₹800 (1). (ii) ₹800 $\times 3$ = ₹2,400 (1). (iii) $8000 \times 1.1^2 = 8000 \times 1.21$ = ₹9,680 (1). (iv) CI $= 1680$, SI $= 1600$ (1); difference ₹80 (1).	1+1+1+2
20	Year 1: A1 = ₹ 33000, interest ₹ 3000 (1); year 2: A2 = ₹ 36300, interest ₹ 3300 (2); year 3: A3 = ₹ 39930, interest ₹ 3630 (1); total ₹ 9930 = A3 − P (1).	1+2+1+1
	OR: Let CP = x. SP = x(1+100/100)(1−40/100) = 1.2x (2); profit = 0.2x = 400 (2); x = ₹ 2000 (1).	2+2+1

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